

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

SURVEYING (CE F213)

Mid Semester Test

Date: 03/10/2016

Duration: 1 Hr. 30 min

Maximum Marks: 30

Type: Closed book

Notes: All questions are compulsory.

Draw figure wherever necessary.

Q.1. Write a note on optical square. [2]

Q.2. What is local attraction? How would you detect the presence of local attraction in an area? [1]

Q.3. What do you understand by contour interval and explain the factors affecting contour interval? [3]

Q.4. Describe the procedure of setting up the plane table over a station? [4]

Q. 5. Fill in the blanks [2]

1. The triangle formed by joining the positions of the known points or their locations on a plane table, is known as _____.
2. Two contours of different elevations do not cross each other except in case of _____.
3. The pull which equalizes the correction due to sag and pull, is known as _____.
4. The BM established by the survey of India is known as _____.
5. In the _____ system, the bearings of lines are measured clockwise or anticlockwise from north or south, whichever is nearer to the line.
6. Map having RF less than $1/500$ is said to be _____ scale.
7. The angle of dip is _____ at the equator.
8. When the parallax angle defined by means of stadia wires is kept _____ and the staff intercept is _____, the method is called as fixed hair method of tacheometry.

Q. 6. The three bays of a base line were measured by a steel tape in catenary as 30.101, 29.739, 26.037 m, under respective pulls of 7, 7, 5 kg, temperature of 14°C , 19°C and 13°C and differences of level of supports of 0.4, 0.7, 0.8 m. If the tape was standardized at a temperature of 15°C under a pull of 4.5 kg, what are the lengths of bays? 30 m of tape is exactly 1 kg with steel at 8300 kg/m^3 . The coefficient of expansion is 0.000011 per $^{\circ}\text{C}$ and the modulus of elasticity $E = 2.1 \times 10^5\text{ N/mm}^2$, cross-sectional area of tape = 0.025 cm^2 . [6]

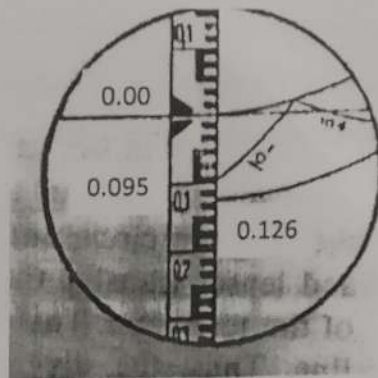
Q.7. The following consecutive readings were taken with a levelling instrument at interval of 20 m.
 2.375, 1.730, 0.615, 3.450, 2.835, 2.070, 1.835, 0.985, 0.435, 1.630, 2.255 and 3.630 m.

The instrument was shifted after the fourth and eighth readings. The last reading was taken on a BM of RL 110.200 m. Find the RLs of all the points using rise and fall method. [4]

Q.8.

[2 * 4]

- A plan drawn to a scale of 1: 4000 was measured by mistake by a scale of 1: 4500. Determine the percentage error in the measured length and the measured area.
- Find out horizontal distance and difference in height in meter using auto reduction telescope shown in the figure below when the telescope is depressed.



- What is the visible horizon distance from a tower 50 m high.? What is the dip of the horizon (in minutes), assuming the radius of the earth to be 6370 km?
- A line AC had the magnetic bearings of $67^{\circ}15'$ in 1960. If the magnetic declination was found to be $21^{\circ}30'E$ by interpolation from an isogonic chart dated 1950 with an annual change of $1'$ westward. Determine the true bearings of line AC.